29/07/2026

Code-B_Phase-1

Aakash
Medical|IIT-JEE|Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

Term Exam for NEET-2027 (XII Studying) Ph-1 - TE02B

Time : 180 Min.

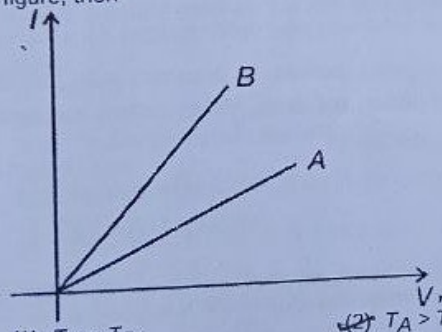
Topics Covered:
Physics: Current Electricity, Moving Charges and Magnetism, Magnetism and Matter
Chemistry: Chemical Kinetics, The d- & f-Block Elements
Botany: Principles of Inheritance & Variation
Zoology: Reproductive Health

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

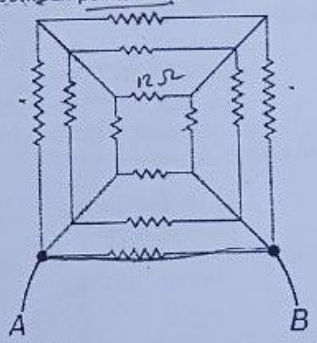
1. For a conductor, graph between current (I) and potential difference (V) at two different temperature is as shown in figure, then



(1) $T_A = T_B$
(2) $T_A > T_B$
(3) $T_B > T_A$
(4) $T_B = 2T_A$

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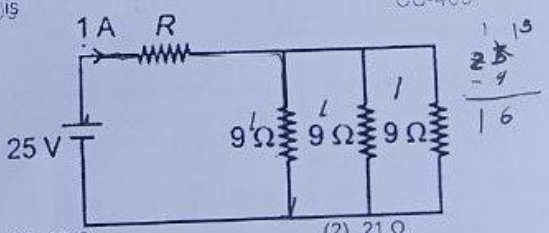
2. Twelve resistors each of resistance 12 Ω are connected in the circuit as shown in the figure. The net resistance between points AB is



(1) 4 Ω
(2) 3 Ω
(3) 2 Ω
(4) 1 Ω

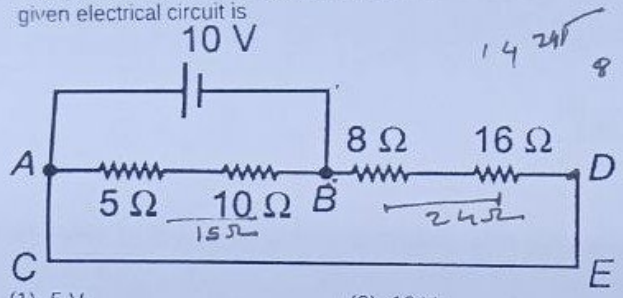
3. In the circuit as shown in the figure, the value of resistance R

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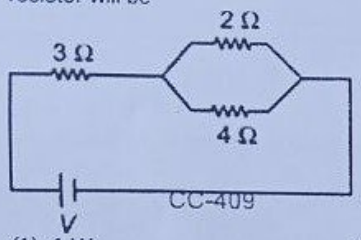
(1) 22 Ω
(2) 21 Ω
(3) 25 Ω
(4) 16 Ω

4. The potential difference across the points B and D in the given electrical circuit is



(1) 5 V (2) 10 V
(3) 6 V (4) 8 V

5. In the following circuit diagram, the power dissipated across 2 Ω resistor is 4 W, then the power dissipated across 4 Ω resistor will be



(1) 1 W (2) 6 W
(3) 4 W (4) 2 W

6. When the current i is flowing through a conductor, the drift velocity is v. If current $2i$ is flowed through the same metal but having double the area of cross-section, then the drift velocity will be

(1) $\frac{v}{4}$ (2) $\frac{v}{2}$
(3) v (4) $4v$

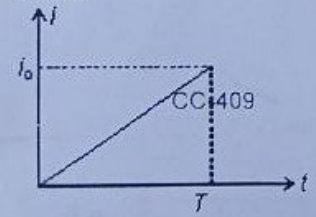
7. A 4 V potential difference is applied across a wire of length 0.2 m, the drift speed of electrons is 1.5×10^{-3} m/s. If the electron density in the wire is 6×10^{28} m^{-3}, the resistivity of the wire is

(1) 1.4×10^{-6} Ωm (2) 2.8×10^{-5} Ωm
(3) 0.14×10^{6} Ωm (4) 0.28×10^{-5} Ωm

8. A steady current flows in a metallic conductor of non-uniform cross-section. The quantity which is constant along the conductor is

(1) Current (2) Current density
(3) Electric field (4) Drift speed

9. The electric current through a wire varies with time as shown.



The total charge flown across any cross-section of the wire in time interval O - T is

(1) i_0T (2) $\frac{i_0T}{\sqrt{2}}$
(3) $\frac{i_0T}{2}$ (4) $\frac{i_0t}{3}$

10. If the conductivity of copper is 6×10^{7} Ω^{-1} m^{-1}, then resistance of a copper wire of length 60 cm and area of cross-section 1 cm^2 is

(1) 0.1 mΩ (2) 10 mΩ
(3) 1 mΩ (4) 100 mΩ

11. If two conducting wires A and B of same dimensions have electron density ratio 1 : 3. If their relaxation time is same then the ratio of resistance of A to resistance of, B is

(1) 1 : 1 (2) 3 : 1
(3) 1 : 3 (4) 9 : 1

12. A wire of length l has a resistance R. Match Column I with Column II.

	Column I		Column II (New resistance)
(A)	If wire is stretched so that its length is four times.	(P)	$16R$
(B)	If wire is stretched so that it length is doubled.	(Q)	R
(C)	If radius of wire is halved keeping length constant.	(R)	$\frac{R}{4}$
(D)	If wire is bent to form L-shaped wire.	(S)	$4R$

(1) A(P), B(Q), C(R), D(S) (2) A(R), B(S), C(Q), D(P)
(3) A(P), B(S), C(S), D(Q) (4) A(S), B(R), C(P), D(Q)

13. A current carring loop consists of 3 identical quarter circle of radius R, lying in the positive quadrants of the x-y, y-z and z-x planes with their centres at the origin, joined together then magnitude of magnetic field at origin is

(1) $\frac{\mu_0 I}{8R}$
(2) $\frac{\sqrt{3}\mu_0 I}{8R}$
(3) $\frac{\mu_0 I}{4R}$
(4) $\frac{\mu_0 I}{\sqrt{3}R}$

14. In a field, a charged particle is projected with speed $\left(2\hat{i}\right)$ m/s and it moves in a circle. Which uniform magnetic field may be present there?

(1) $\left(2\hat{j}\right)$ T (2) $\left(4\hat{i}\right)$ T
(3) $\left(8\hat{k}\right)$ T (4) Both (1) and (3)

15. The value of shunt which allows only 20% of main current through a galvanometer of resistance 80 Ω is

(1) 40 Ω (2) 16 Ω
(3) 20 Ω (4) 320 Ω

16. A charged particle is projected in to a region where there may be an electric filed $\vec{E}$ and/or magnetic field $\vec{B}$. If the charged particle goes un-accelerated, then it is not possible that

(1) $\vec{E} \neq 0, \vec{B} = 0$ (2) $\vec{E} = 0, \vec{B} = 0$
(3) $\vec{E} = 0, \vec{B} \neq 0$ (4) $\vec{E} \neq 0, \vec{B} \neq 0$

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17. Consider the following statements:
Statement (A): Biot-Savart's law provides the expression for the magnetic field generated by an infinitesimal current element of a current carrying conductor.
Statement (B): Biot-Savart's law is similar to Coulomb's inverse square law for electric charges. However, while Coulomb's law involves a scalar source (charge q), Biot-Savart's law involves a vector source (current element $Id\vec{l}$).
In light of the above statements choose the most appropriate answer from the options given below.
(1) Statement (A) is correct and statement (B) is incorrect.
(2) Statement (A) is incorrect and statement (B) is correct.
(3) Both statements (A) and (B) are correct.
(4) Both statements (A) and (B) are incorrect.

18. A particle of charge Q and mass m moves in a circular orbit of radius R with angular speed ω. The ratio of magnitude of its magnetic moment to that of its angular momentum is
(1) $\frac{Q}{m}$ (2) $\frac{Q}{2m}$
(3) $\frac{2Q}{m}$ (4) $\sqrt{\frac{2Q}{m}}$

19. A loop of magnetic moment M is placed in the orientation of stable equilibrium position in a uniform magnetic field B then the external work done in rotating it through an angle θ is
(1) $MB\cos\theta$ (2) $-MB\cos\theta$
(3) $-MB(1-\cos\theta)$ (4) $MB(1-\cos\theta)$

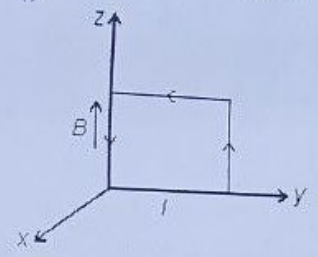
20. In the relation $\vec{F} = q\left(\vec{V} \times \vec{B}\right)$, which pairs is/are always perpendicular to each other?
(1) F and V (2) F and B
(3) V and B (4) Both (1) and (2)

21. The magnetic field due to a current carrying circular loop of radius 3 m at a point on the axis at a distance of 4 m from the centre is 54 μT. What will be its value at the centre of the loop?
(1) 250 μT (2) 150 μT
(3) 125 μT (4) 75 μT

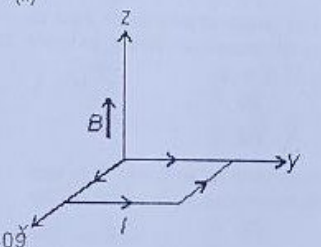
22. A thin walled pipe of infinite length carries a current along its length, then
(1) The magnetic field is different at different points inside the pipe
(2) The magnetic field is non-zero and same at all points inside the pipe
(3) The magnetic field at any point inside the pipe is zero
(4) The magnetic field is zero only on the axis of the pipe

23. If a charged particle is thrown perpendicular to a region of crossed electric field (E) and magnetic field (B) goes undeviated, then the speed of particle is
(1) $\frac{B}{E}$ (2) $\frac{E}{B}$
(3) $\frac{B^2}{E^2}$ (4) $\frac{E^2}{B^2}$

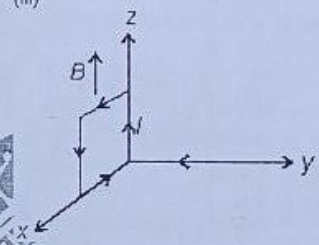
24. A rectangular loop carrying a current I is placed in different orientations as shown in figures below
(i)



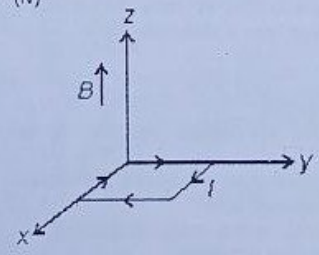
(ii)



(iii)



(iv)



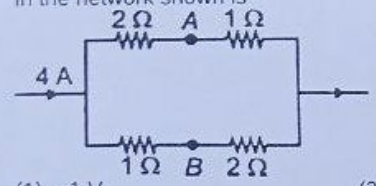
If there is unlform magnetic field in the +z-direction, in which orientations the loop would be in (a) stable equilibrium, (b) unstable equilibrium respectively
(1) (i) and (iii) (2) (ii) and (iv)
(3) (ii) and (iii) (4) (i) and (ii)

25. Two long wires one carrying current 1 A and other carrying 2 A in the same direction are placed parallel to each other produces a net magnetic field B at the mid-point. What will be the field at mid-point when current 1 A is switched off?
(1) B (2) $2B$
(3) $\frac{B}{2}$ (4) $\frac{3B}{2}$

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26. Susceptibility of a ferromagnetic substance is
(1) Positive and large
(2) Negative and large
(3) Zero
(4) May be positive or negative

27. The potential difference ($V_A - V_B$) between points A and B in the network shown is



(1) −1 V (2) −2 V
(3) −4 V (4) −3 V

28. A magnetic wire is turned into shape of semicircle then its magnetic moment becomes M, magnetic dipole moment before turning was
(1) $2M\pi$ (2) $\frac{M\pi}{2}$
(3) $\frac{M\pi}{\sqrt{2}}$ (4) $\frac{M}{2}$

29. A steel wire of length l has magnetic moment M, it is bent into L shape from middle. The new magnetic moment will be
(1) $\frac{M}{2}$ (2) $\frac{M}{\sqrt{2}}$
(3) $\frac{M}{3}$ (4) $\sqrt{2}M$

30. SI unit of quantity represented by area of Hysteresis loop is
(1) $\frac{J}{m^3}$ (2) $\frac{A^2}{m^2}$
(3) $\frac{J}{m^2}$ (4) $\frac{A}{m^2}$

31. A paramagnetic substance of susceptibility 3×10^{-4} is placed in a magnetising field of 4×10^{-4} A m^{-1}. Then the intensity of magnetisation in A m^{-1} is
(1) 0.75×10^{-8} (2) 1.2×10^{-7}
(3) 1.33×10^{-8} (4) 14×10^{-7}

32. Ratio of magnitude of magnetic field at same distance on axial line and on equatorial line from a short magnet is
(1) 1 : 1 (2) 2 : 1
(3) 4 : 1 (4) 8 : 1

33. Consider the following statements:
(A) Diamagnetic substances are weakly repelled by a magnetic field.
(B) For a paramagnetic material, magnetisation is always in the direction opposite to the applied field.
(C) Ferromagnetic substances retain magnetisation even after the external magnetic field is removed.
The correct statement(s) is/are
(1) Only (A) (2) Only (B)
(3) Both (A) and (B) (4) Both (A) and (C)

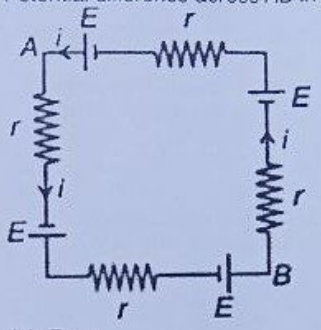
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34. A magnetic material of volume 30 cm^3 is placed in a magnetising field. The magnetic moment produced due to it is 6 A m^2. The magnetisation of the sample is
(1) 2×10^5 A m^{-1} (2) 10^5 A m^{-1}
(3) 10^4 A m^{-1} (4) 2×10^{-5} A m^{-1}

35. The ratio of intensity of magnetisation to the magnetising field is called
(1) Magnetic intensity (2) Magnetic susceptibility
(3) Magnetic permeability (4) Magnetic induction

36. Through a given cross section n_1 electrons per second are passing from left to right and n_2 protons per second are passing from right to left simultaneously. The electric current through that cross section is (e = electronic charge)
(1) $(n_1 + n_2)e$ toward left (2) $(n_2 - n_1)e$ toward right
(3) $(n_1^2 + n_2^2)e$ toward left (4) $(n_2^2 - n_1^2)e$ toward right

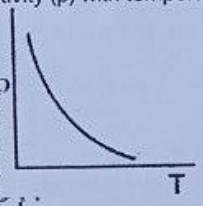
37. Potential difference across AB in the network shown is

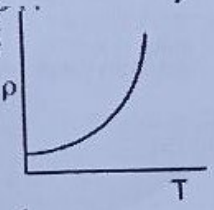


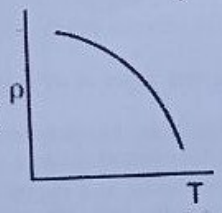
(1) Zero (2) E
(3) $E - \frac{i}{2}$ (4) $E - 2ir$

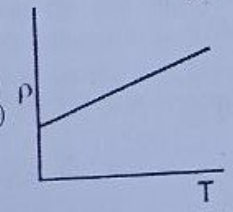
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38. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

(1)


(2)


(3)


(4)


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39. Which one of the following has negative temperature coefficient of resistance?
(1) Copper (2) Silver
(3) Gold (4) Germanium

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40. A particle with a specific charge(s) is fired with a speed v towards a wall at a distance d, perpendicular to the wall. What minimum magnetic field must exist in this region for the particle doesn't hit the wall?
(1) $\frac{v}{sd}$ (2) $\frac{2v}{sd}$
(3) $\frac{v}{2sd}$ (4) $\frac{v}{4sd}$

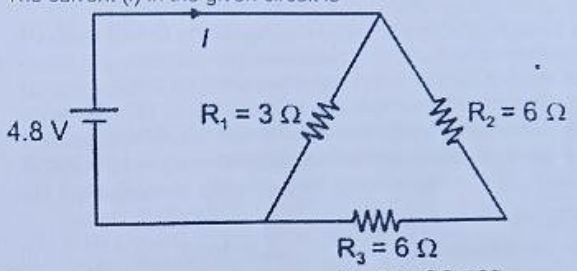
41. To make the magnetic field radial in a moving coil galvanometer
(1) The number of turns in the coil is increased
(2) The number of turns in coil is decreased
(3) Poles are cylindrically cut
(4) Coil is wound on aluminium frame

42. In a toroid the number of turns per unit length is 1000 and current through it is $\frac{1}{4\pi}$ ampere. The magnetic field produced inside (in weber/m^2) will be
(1) 10^{-2} (2) 10^{-3}
(3) 10^{-4} (4) 10^{-7}

43. Choose the incorrect statement.
(1) Magnetic moment is a vector quantity.
(2) Geometrical length of a bar magnet is less than magnetic length.
(3) Magnetic field lines inside a bar magnet is from south pole to north pole.
(4) Pole strength is a scalar quantity.

44. Bohr magneton is given by (symbols have their usual meanings)
(1) $\frac{4\pi m_e}{eh^2}$ (2) $\frac{4\pi m_e}{eh}$
(3) $\frac{eh^2}{4\pi m_e}$ (4) $\frac{eh}{4\pi m_e}$

45. The current (I) in the given circuit is



(1) 1.6 A (2) 2 A
(3) 0.32 A (4) 3.2 A

CHEMISTRY

46. Given below are two statements
Statement I : Rate of reaction does not depend on concentration of reactant for a zero order reaction.
Statement II : Decomposition of $NH_3(g)$ on a hot Pt(s) surface at high pressure is zero order reaction
In light of the above statements, choose the most appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

47. If the rate constant of a reaction is 6.93×10^{-2} min^{-1} then half-life period of the reaction is
(1) 15 minute (2) 10 minute
(3) 20 minute (4) 5 minute

48. The experimental data for a reaction $2X + Y \rightarrow X_2Y$ is

Exp.	[X] M	[Y] M	rate (Ms^{-1})
1	0.1	0.2	1.2×10^{-3}
2	0.1	0.1	3×10^{-4}
3	0.2	0.2	2.4×10^{-3}

The rate equation for above data is
(1) $r = k[X]^2[Y]$ (2) $r = k[X][Y]^2$
(3) $r = k[X][Y]$ (4) $r = k[X]^2[Y]^2$

49. For the reaction, $A(g) + 2B(g) \rightarrow 4C(g) + 3D(g)$ The rate of reaction can be expressed by
(1) $\frac{4d[C]}{dt}$ (2) $-\frac{1}{3}\frac{d[D]}{dt}$
(3) $\frac{d[A]}{dt}$ (4) $-\frac{1}{2}\frac{d[B]}{dt}$

50. For the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the rate of change of concentration for hydrogen is 4 g/h. The rate of change of concentration of NH_3 is
(1) 2.6 g/h (2) 22.7 g/h
(3) 45.33 g/h (4) 34 g/h

51. The rate constant of substrate A for a reaction is 2×10^{-3} s^{-1}. If the rate of reaction is 7×10^{-4} mol L^{-1} s^{-1} then the concentration of substrate A (in mol L^{-1}) is
(1) 1.5 M (2) 0.8 M
(3) 0.35 M (4) 2.25 M

52. Rate of reaction gets doubled on increase of every 10°C rise of temperature. If temperature is increased from 30°C to 60°C, the rate of reaction becomes
(1) 6 times (2) 30 times
(3) 8 times (4) 3 times

53. Match times in list I with their values in list II for first order reaction

	List I		List II
a.	$t_{1/2}$	(i)	$\frac{\ln 4}{k}$
b.	$t_{75\%}$	(ii)	$\frac{\ln 2}{k}$
c.	$t_{99\%}$	(iii)	$\frac{\ln 10}{k}$
d.	$t_{90\%}$	(iv)	$\frac{2 \ln 10}{k}$

Correct match is
(1) a(ii), b(i), c(iii), d(iv) (2) a(i), b(ii), c(iv), d(iii)
(3) a(ii), b(i), c(iv), d(iii) (4) a(ii), b(iii), c(iv), d(i)

54. If rate constant and rate of reaction both have same units, then the order of reaction is
(1) 0 (2) 1
(3) 2 (4) 3

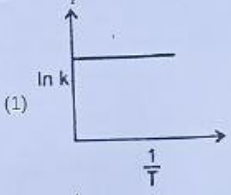
55. Given below are two statements:
Statement I: Inversion of cane sugar in acidic medium is pseudo first order reaction.
Statement II: Molecularity of a reaction can be zero.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct

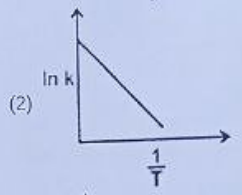
56. Given below are two statements
Statement I : A catalyst provides an alternate pathway or reaction mechanism by reducing the activation energy between reactants and products.
Statement II : Catalyst catalyses the spontaneous reactions but does not catalyse non-spontaneous reactions.
In the light of above statements, choose the correct option.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

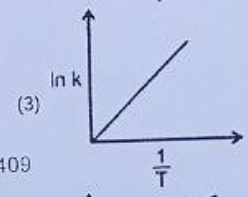
57. For given reaction activation energy of forward direction is 40 kJ mol^{-1} and for backward direction is 60 kJ mol^{-1} then $\Delta_r H$ for reaction A $\rightarrow$B will be
(1) -20 kJ mol^{-1} (2) 20 kJ mol^{-1}
(3) 100 kJ mol^{-1} (4) -100 kJ mol^{-1}

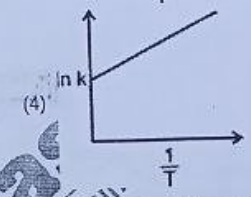
58. If rate constant 'k' of a reaction varies with temperature as $\ln k = 2 - \frac{200}{T}$, then Arrhenius constant (A) and activation energy (E_a) of the reaction respectively are
(1) 100, 200 R (2) 100, 20 R
(3) e^2, 200 R (4) e^2, 20 R

59. According to Arrhenius equation, which is correct plot of $\ln k$ vs $\frac{1}{T}$?

(1) 

(2) 

(3) 

(4) 

60. The rate constant of a reaction is 1.5×10^7 s^{-1} at 50°C and 4.5×10^7 s^{-1} at 100°C. What is the value of activation energy?
(1) 2.2×10^3 J mol^{-1} (2) 2300 J mol^{-1}
(3) 2.2×10^4 J mol^{-1} (4) 220 J mol^{-1}

61. Consider the following statements
(a) The oxidation states of Chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same.
(b) The stability of Cu^{2+} is more than Cu^+ salts in aqueous solution.
(c) The catalytic activity of transition metals and their compounds is ascribed to their ability to adopt variable oxidation states
The correct statements is/are
(1) (b) and (c) only (2) (a) and (c) only
(3) (b) only (4) (c) only

62. Which among the following transition elements has positive $E^\circ_{M^{2+}/M}$ value ?
(1) Ti (2) V
(3) Cu (4) Cr

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63. Match List-I with List-II.

	List-I (Catalyst)		List-II (Process)
(a)	$TiCl_4$ + $Al(CH_3)_3$	(i)	Haber's process
(b)	V_2O_5	(ii)	Polymerisation of alkynes
(c)	Ni complex	(iii)	Oxidation of SO_2 in the manufacture of H_2SO_4
(d)	Iron catalyst	(iv)	Polymerisation of Ethylene

Choose the **correct** option.
(1) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)
(2) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
(3) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)
(4) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

64. Given below are some aquated transition metal ions
(a) Sc^{3+}
(b) Ti^{4+}
(c) Cr^{3+}
(d) Mn^{2+}
which of them are colourless.
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (d) only (4) (a), (b) and (c) only

65. First Ionisation enthalpy is maximum for
(1) Cu (2) Zn
(3) Cr (4) Mn

66. Consider the following statements
(a) Zinc has minimum value of enthalpy of atomisation in 3d series
(b) Mn_2O_7 is a covalent green oil
(c) In acidic medium, MnO_4^{2-} disproportionates into MnO_4^- and MnO_2
The **correct** statement(s) is/are
(1) (a), (b) and (c) (2) (a) only
(3) (b) and (c) only (4) (a) and (b) only

67. Correct order of negative hydration enthalpy of V^{2+}, Co^{2+}, Fe^{2+} and Cr^{2+} is
(1) $V^{2+} > Co^{2+} > Cr^{2+} > Fe^{2+}$
(2) $Co^{2+} > Fe^{2+} > Cr^{2+} > V^{2+}$
(3) $Fe^{2+} > Co^{2+} > Cr^{2+} > V^{2+}$
(4) $Co^{2+} > Cr^{2+} > Fe^{2+} > V^{2+}$

68. Consider the following statements about interstitial compounds
(i) They are chemically inert
(ii) They are very hard
(iii) They retain metallic conductivity
The correct statements are
(1) (i) and (ii) only (2) (ii) and (iii) only
(3) (i) and (iii) only (4) (i), (ii) and (iii)

69. The metal which will not liberate H_2 gas on reaction with dilute HCl is
(1) Mn (2) Cu
(3) Cr (4) Ti

70. Which among the following has maximum density?
(1) Fe (2) Ti
(3) Ni (4) Cr

71. The correct order of basicity of the given lanthanoid hydroxides is
(1) $Nd(OH)_3 > Eu(OH)_3 > Gd(OH)_3 > Pr(OH)_3$
(2) $Pr(OH)_3 > Nd(OH)_3 > Eu(OH)_3 > Gd(OH)_3$
(3) $Eu(OH)_3 > Gd(OH)_3 > Pr(OH)_3 > Nd(OH)_3$
(4) $Gd(OH)_3 > Eu(OH)_3 > Nd(OH)_3 > Pr(OH)_3$

72. The number of mole of $KMnO_4$ that will be needed to react with two moles of sulphite ion in acidic medium is
(1) 1/5 (2) 3/5
(3) 4/5 (4) 2/5

73. Identify the correct electronic configuration of Gadolinium in ground state.
(1) $[Xe]\,4f^8 6s^2$ (2) $[Xe]\,4f^7 5d^1 6s^2$
(3) $[Xe]\,4f^9 6s^1$ (4) $[Xe]\,4f^7 6s^2$

74. Given below are two statements: one is labelled as Assertion (A) and other is labelled as a Reason (R)
Assertion (A): Manganate ion is paramagnetic in nature.
Reason (R): Manganate ion contains one unpaired electron.
In light of above statements, choose the correct answer.
(1) Both (A) and (R) are true statements and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true statements but (R) is not the correct explanation of (A)
(3) (A) is false but (R) is true
(4) (A) is true but (R) is false

75. Which of the following lanthanoids exhibit +4 oxidation state?
(1) Ce (2) Eu
(3) Sm (4) Yb

76. Mischmetall contains maximum percent of
(1) Lanthanoid (2) Carbon
(3) Calcium (4) Aluminium

77. Which of the following ions is smallest in size?
(1) La^{3+} (2) Nd^{3+}
(3) Ce^{3+} (4) Dy^{3+}

78. Number of Cr–O–Cr bond(s) present in dichromate ion is
(1) 1 (2) 4
(3) 2 (4) Zero

79. Which among the following actinoids exists in +4 oxidation state only?
(1) Ac (2) Th
(3) U (4) Np

80. The correct electronic configuration of Curium (atomic number = 96) is
(1) $[Rn]5f^9 6s^1$ (2) $[Rn]5f^6 6d^2 7s^2$
(3) $[Rn]5f^7 6d^1 7s^2$ (4) $[Rn]5f^8 6s^2$

81. A gaseous reaction,
$A_2(g) \rightarrow B(g) + \frac{1}{2}C(g)$
shows increase in pressure from 100 mmHg to 120 mmHg in 5 minutes. The rate of disappearance of A_2 is
(1) 4 mmHg min^{-1} (2) 8 mmHg min^{-1}
(3) 16 mmHg min^{-1} (4) 2 mmHg min^{-1}

82. For the homogeneous elementary reaction, $A + B \rightarrow C$, the unit of rate constant is
(1) s^{-1} (2) s^{-1} mol L^{-1}
(3) s^{-1} mol^{-1} L (4) s

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83. The rate law for the reaction,
$RCl + NaOH(aq) \rightarrow ROH + NaCl$
is given by Rate = k[RCl]. The rate of reaction will be
(1) Doubled on doubling the concentration of NaOH
(2) Halved on reducing the concentration of alkyl halide to one half
(3) Decreases on increasing the temperature
(4) Unaffected by increasing the temperature

84. A sample of a radioactive substance undergoes 80% decomposition in 345 minutes. The half life of reaction in minutes will be
(1) $\frac{\ln 2}{\ln 5} \times 345$ (2) $\frac{\ln 5}{\ln 2} \times 345$
(3) $\frac{\ln 5}{\ln 4} \times 345$ (4) $\frac{\ln 4}{\ln 5} \times 345$

85. A substance (initial concentration 'a') reacts according to zero order kinetics. The time it takes for the completion of the reaction is
(1) $\frac{a}{k}$ (2) $\frac{a}{2k}$
(3) $\frac{k}{a}$ (4) $\frac{2k}{a}$

86. Gunmetal contains
(1) Cu, Sn, Zn (2) Cu, Ni
(3) Cu, Ni, Fe (4) Cu, Sn, P

87. The blue colour produced on adding H_2O_2 to acidified $K_2Cr_2O_7$ is due to the formation of
(1) CrO_5 (2) Cr_2O_3
(3) CrO_4^{2-} (4) CrO_3

88. The element which does not show d^0 configuration in its highest oxidation state
(1) V (2) Mn
(3) Cr (4) Fe

89. Gold dissolves in aqua regia to give
(1) $H[AuCl_4]$ (2) $AuNO_3$
(3) $H_2[AuCl_6]$ (4) $Au(NO_3)_3$

90. Transuranic elements begin with
(1) Np (2) Cm
(3) Pu (4) U

BOTANY

91. Chromosomal complement found in patients afflicted with Klinefelter's syndrome is
(1) 44 + XO (2) 44 + XY
(3) 46 + XX (4) 44 + XXY

92. In any cross, if recombination frequency is 15%, it means
(1) The distance between the observed genes is 15 cM
(2) Number of recombinants are 15 in 20 progenies
(3) Number of parental types are 15 in 20 progenies
(4) The genes are tightly linked.

93. In garden pea, gene controlling starch synthesis is related to all of the given phenomenon, **except**
(1) Pleiotropy (2) Incomplete dominance
(3) Codominance (4) Complete dominance

94. Regarding blood group, the genotypes of a husband and his wife are I^AI^B and I^Ai. They have three children. The blood group not possible in their children is
(1) A (2) B
(3) O (4) AB

95. Match **List I** with **List II**.

	List I		List II
a.	Incomplete dominance	(i)	Phenylketonuria
b.	Codominance	(ii)	Myotonic dystrophy
c.	Autosomal dominant trait	(iii)	Phenotype that did not resemble either of the two parents and was in between the two
d.	Pleiotropy	(iv)	F_1 generation resembles both parents

Select the **correct** answer from the following options.
(1) a(iv), b(i), c(iii), d(ii) (2) a(iii), b(iv), c(ii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(ii), c(i), d(iv)

96. Haplodiploid method of sex determination is found in
(1) Butterfly (2) Honey bee
(3) Grasshopper (4) Fruit fly

97. The term used for the units of inheritance by Mendel was
(1) Traits (2) Genes
(3) Alleles (4) Factors

98. Mark True **(T)** or False **(F)** for the given statements and select the **correct** option.
(A) Mendel selected only 7 true-breeding pea plant varieties.
(B) His experiments had a small sampling size which gave greater credibility to the data that he collected.
(C) Mendel for the first time applied statistical analysis and mathematical logic to problems in biology.

	A	B	C
(1)	T	T	F
(2)	T	F	T
(3)	F	T	T
(4)	F	F	T

(1) (1) (2) (2)
(3) (3) (4) (4)

99. α-Thalassemia is
(1) Caused due to abnormality in chromosome 11
(2) Commonly observed in females only
(3) Controlled by two closely linked genes on chromosome 16 of each parent
(4) Controlled by HBA1 gene and HBB gene

100. Which among the given is **not** a dominant trait of pea?
(1) Green pod (2) Violet flower
(3) Round seed (4) Constricted pod

101. Read the following statements and select the **correct** option.
Statement A : Punnett square is a graphical representation only to calculate the probability of all possible phenotypes of offspring in a genetic cross.
Statement B : The symbols ♀ and ♂ are used to denote female and male, respectively.
(1) Only statement A is true
(2) Only statement B is true
(3) Both the statements A and B are true
(4) Both the statements A and B are false

102. Point mutation is a change in
(1) A single base pair of DNA
(2) Entire sequence of base pairs in mRNA
(3) Three base pairs of DNA due to deletion
(4) Two base pairs of DNA due to addition

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103. Which of the given is not a test cross?
(1) Tt × tt
(2) RrYy × rryy
(3) $\frac{w^+ \; y^+}{w \; y} \times \frac{w \; y}{w \; y}$
(4) rryy × rryy

104. Fruit flies are suitable experimental organisms for genetic studies because of all of the following, **except**
(1) They could be grown on simple synthetic medium in the laboratory
(2) They have short life span
(3) They produce large number of progenies in each mating
(4) The male and female flies are indistinguishable

105. Slightly different forms of the same gene are called
(1) Alleles
(2) Polygenes
(3) Pleiotropic gene
(4) Chromosomes

106. How many types of gametes will be produced by individuals of AaBBCc genotype?
(1) Four
(2) Two
(3) Six
(4) Eight

107. Match column-I with column-II

Column I	Column II
a. Turner's syndrome	(i) Broad palm with characteristic palm crease
b. Down's syndrome	(ii) Absence of one X chromosome
c. Klinefelter's syndrome	(iii) Inborn error of metabolism
d. Phenylketonuria	(iv) Gynaecomastia

Select the **correct** option from the following:
(1) a(ii); b(iii); c(i); d(iv)
(2) a(ii); b(i); c(iv); d(iii)
(3) a(ii); b(iv); c(iii); d(i)
(4) a(i); b(ii); c(iv); d(iii)

108. Red eye colour of *Drosophila* is
(1) A recessive trait
(2) Regulated by a gene present on Y-chromosome
(3) A sex-linked trait
(4) Seen in females only

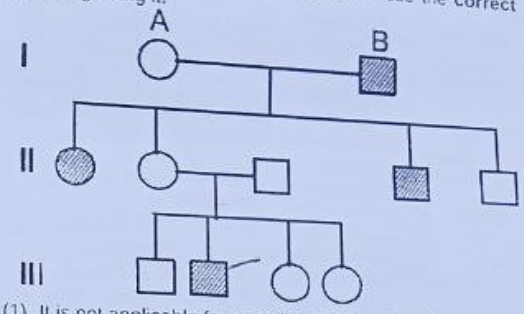
109. According to 'Law of segregation'
(1) Two factors of a character get mixed up in F_2 generation
(2) During gamete formation, factors of a character get mixed up
(3) Factors of a character separate from each other during gamete formation
(4) Out of two factors for each character, dominant will be able to express in F_1 generation

110. Phenotype is
(1) Genetic constitution of an organism
(2) Observable morphological appearance
(3) Homozygosity of the organism
(4) Ability of an organism to form gametes

111. A normal man whose father was haemophilic, married a normal woman whose mother was haemophilic. What are the chances of getting haemophilic offsprings?
(1) 75%
(2) 50%
(3) 25%
(4) 100%

112. Chromosomal aberration includes
(1) Deletion of a part of a chromosome
(2) Transition
(3) Frame shift mutation
(4) Deletion of two bases

113. Study the following pedigree chart and choose the **correct** option regarding it.



(1) It is not applicable for an X-linked recessive disorder
(2) The genotype of A can be X^CX
(3) It shows inheritance of a Y-linked disorder
(4) The genotype of B cannot be aa

114. Read the following assertion (A) and reason (R) and choose the **correct** answer.
Assertion (A): In phenylketonuria affected individual, phenylalanine is accumulated and converted into phenylpyruvic acid.
Reason (R): Individual inflicted with this disorder lacks a liver enzyme called phenylalanine hydroxylase that converts tyrosine into phenylalanine.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

115. Queen Victoria was carrier of the disease
(1) Sickle–cell anaemia
(2) Phenylketonuria
(3) Myotonic dystrophy
(4) Haemophilia

116. Select the **correct** option w.r.t. law of independent assortment
(1) It can be explained by using monohybrid cross
(2) Inheritance of one character is dependent on another character
(3) This law is not applicable universally
(4) It was proposed by Bateson

117. Which one is **incorrect** regarding chromosomal theory of inheritance?
(1) A gamete contains only one chromosome and only one allele
(2) The paired condition of homologous chromosomes as well as Mendelian factors is restored during fertilization
(3) Chromosomes as well as genes occur in pairs in the diploid cells
(4) Homologous chromosomes as well as genes segregate at the time of gamete formation

118. A pink flowered plant of snapdragon was crossed with red flowered plant. The progeny will have
(1) 1 : 1 ratio of pink and white flowered plants
(2) 1 : 2 : 1 ratio of red, pink and white flowered plants
(3) 1 : 1 ratio of red and pink flowered plants
(4) 100% pink flowered plants

119. Which of the following represents trisomy?
(1) 2n – 2
(2) 2n + 1
(3) 2n + 3
(4) n + 1

120. Match Column-I with Column-II and select the **correct** option.

Column-I		Column-II
(I) Henking	(a)	Coined the term linkage to describe physical association of genes on a chromosome
(II) Alfred Sturtevant	(b)	Traced a specific nuclear structure all through spermatogenesis in a few insects.
(III) T.H. Morgan	(c)	Rediscovered Mendel's results on the inheritance of characters.
(IV) von Tschermak	(d)	Used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes

(1) (I)-(a), (II)-(d), (III)-(c), (IV)-(b)
(2) (I)-(b), (II)-(d), (III)-(a), (IV)-(c)
(3) (I)-(b), (II)-(a), (III)-(d), (IV)-(c)
(4) (I)-(c), (II)-(b), (III)-(a), (IV)-(d)

121. How many phenotypes will be obtained, if a plant heterozygous for two features is crossed with a plant recessive for the same features?
(1) 1 (2) 2
(3) 3 (4) 4

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122. The recessive allele
(1) May produce non functional enzyme
(2) Is always unmodified allele
(3) Always produces normal enzyme
(4) Can express itself in heterozygous condition

123. The degree by which progeny differ from their parents is called
(1) Inheritance (2) Variation
(3) Hybridisation (4) Pollination

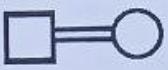
124. Probability of three sons to a couple is
(1) $\frac{1}{2}$ (2) $\frac{1}{4}$
(3) $\frac{1}{8}$ (4) $\frac{1}{16}$

125. A pea plant heterozygous for pod shape and flower colour was selfed and total 512 seeds were obtained. What will be the total number of seeds having the trait for inflated pod and white coloured flowers?
(1) 128 (2) 96
(3) 210 (4) 250

126. F_1 progeny of Mendelian dihybrid cross produces
(1) Two types of pollen grains
(2) Four genotypes of gametes
(3) Two types of eggs
(4) Four types of pollens only

127. Individuals having homomorphic sex chromosomes produce
(1) Only one gamete in complete life span
(2) One type of gametes
(3) No gametes CC-409
(4) Two type of gametes

128. Which of the following phenomena leads to variation in DNA?
(1) Linkage, mutation (2) Recombination, linkage
(3) Mutation, recombination (4) Aneuploidy, linkage

129. In pedigree analysis, symbol given for sex unspecified is
(1) 
(2) 
(3) 
(4) 

130. Phenylketonuria is inherited as
(1) Autosomal recessive trait (2) Sex-linked dominant trait
(3) X-linked recessive trait (4) Autosomal dominant trait

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131. Select the **incorrect** statement w.r.t. 'Haemophilia'
(1) Is an autosomal recessive disorder.
(2) Shows its transmission from unaffected carrier female to some of the male progeny.
(3) In this disease, a single protein that is a part of the cascade of proteins involved in clotting of blood is affected.
(4) The possibility of female becoming a haemophilic is extremely rare.

132. Down's syndrome results in the gain of extra copy of
(1) Chromosome 11 (2) Chromosome 12
(3) Chromosome 21 (4) Chromosome 6

133. Find out the frequency of AabbCcDdee if parents are AabbCcddEe and AabbccDdee
(1) 0.78% (2) 12.5%
(3) 25% (4) 50%

134. Which of the following combinations seems to have some linkages in character selected by Mendel?
(1) Stem height and pod colour
(2) Flower colour and flower position
(3) Seed shape and seed colour
(4) Plant height and pod shape

135. Gregor Mendel, conducted hybridisation experiments on garden pea during the year
(1) 1856-1863 (2) 1768-1775
(3) 1907-1914 (4) 1707-1714

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ZOOLOGY

136. The contraceptive 'SAHELI'
(1) Blocks estrogen receptors in the uterus, preventing eggs from getting implanted
(2) Increases the concentration of estrogen and prevents ovulation in females
(3) Is a post-coital contraceptive
(4) Is an IUD

137. The function of copper ions in copper releasing IUD's is :
(1) They suppress sperm motility and fertilising capacity of sperms
(2) They inhibit gametogenesis
(3) They make uterus unsuitable for implantation
(4) They inhibit ovulation

138. Read the following statements:
Statement (A): Amniocentesis enables pre-natal diagnosis of chromosomal abnormalities and genetic disorders through analysis of foetal cells present in the amniotic fluid.
Statement (B): Prior to statutory ban in India, unchecked misuse of amniocentesis adversely affected sex-ratio, despite its medical utility.
Choose the **correct** option.
(1) Both statements (A) and (B) are incorrect
(2) Both statements (A) and (B) are correct
(3) Only statement (A) is correct
(4) Only statement (B) is correct.

139. Select the option representing the correct pair of hormone releasing intra uterine devices.
(1) Vaults, LNG-20
(2) Multiload 375, Progestasert
(3) Progestasert, LNG-20
(4) Lippes loop, Multiload 375

140. When were the 'family planning' programmes initiated in India?
(1) 1951
(2) 1972
(3) 1945
(4) 1961

141. All of the following can be effective methods to prevent the spread of HIV infection, **except**
(1) Use of only disposable needles and syringes in public and private hospitals and clinics
(2) Free distribution of condoms
(3) Controlling intravenous drug abuse
(4) Advocating unsafe sexual activity

142. Select the **correct** option w.r.t. the full form of GIFT.
(1) Germ cell intra-fallopian transfer
(2) Gamete inter-fallopian transfer
(3) Gamete intra-fallopian transfer
(4) Gamete inter-fertilization transfer

143. Match column I and column II w.r.t. contraceptives and their mode of action. Choose the **correct** answer.

	Column I		Column II
a.	Tubectomy	(i)	Transport of ova through fallopian tube is blocked
b.	Condoms	(ii)	Prevents fertilisation
c.	Copper-T	(iii)	Increases phagocytosis of sperms, suppresses sperm motility and fertilizing capacity of sperms
d.	Hormonal IUD	(iv)	Prevents/retards entry of sperms, and implantation

(1) a(iv), b(iii), c(i), d(ii)
(2) a(iii), b(iv), c(ii), d(i)
(3) a(iv), b(i), c(iii), d(ii)
(4) a(i), b(ii), c(iii), d(iv)

144. The contraceptive device inserted by the doctors or expert nurses in the uterus through vagina is
(1) Implant
(2) IUD
(3) Cervical cap
(4) Diaphragm

145. "Reproductive health means a total well-being in all aspects of reproduction, *i.e.*, physical, emotional, behavioural and social". This definition was given by
(1) GEAC
(2) WHO
(3) CDRI
(4) EFB

146. Implants used as female contraceptives mostly contain the hormone
(1) Parathormone
(2) Progestin
(3) Pitocin
(4) Prolactin

147. All of the following statements are true w.r.t. contraceptive methods, **except**
(1) They are used to prevent/delay pregnancy.
(2) These are used against the natural process of reproduction.
(3) They do not have any side effects at all.
(4) They play important role in preventing the uncontrolled growth of human population.

148. How many of the following is/are method(s) of contraception for males?
a. Condoms
b. Diaphragm
c. Lactational amenorrhea
d. Implants
e. Vasectomy
Select the **correct** option.
(1) Two
(2) Three
(3) One
(4) Five

149. Select the **mismatch** w.r.t. function of contraceptives.

(1)	Oral contraceptive pills	–	Prevent ovulation
(2)	Barrier methods	–	Prevent physical meeting of gametes
(3)	Lactational amenorrhoea	–	Prevents lactation
(4)	Vasectomy	–	Prevents gamete transport

(1) (1)
(2) (2)
(3) (3)
(4) (4)

150. Which of the following is **not** a copper releasing IUD?
(1) LNG-20
(2) CuT
(3) Cu7
(4) Multiload 375

151. The withdrawal method of contraception is
(1) Also called the 'safe period' method
(2) Not a natural method of contraception
(3) Characterized by many side-effects
(4) Also called coitus interruptus

152. Steroidal pills have to be taken daily for a period of
(1) 21 days starting preferably within the first five days of menstrual cycle
(2) 14 days starting preferably within the first five days of menstrual cycle
(3) 14 days starting preferably within the first seven days of menstrual cycle
(4) 21 days starting preferably within the last seven days of menstrual cycle

153. Method of contraception in which the human females track their menstrual history to predict their day of ovulation based on which they avoid coitus, is
(1) Rhythm method
(2) Lactational amenorrhea
(3) Chemical contraception
(4) Surgical method of contraception

154. If we consider GIFT and ZIFT, fertilization in former and latter occur
(1) *in vitro* and *in vivo*, respectively
(2) *in vivo* and *in vitro*, respectively
(3) *in vivo* and *in vivo*, respectively
(4) *in vitro* and *in vitro*, respectively

155. Intra uterine transfer technique in ART involves transfer of
(1) Sperms into uterus
(2) Embryo with more than 8 blastomeres into uterus
(3) Mixture of sperm and ovum into uterus
(4) Mixture of sperm and ovum into ovary

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156. Which one of the following groups include all sexually transmitted diseases?
(1) AIDS, Syphilis, Cholera
(2) HIV, Malaria, Trichomoniasis
(3) Gonorrhoea, AIDS, Chlamydiasis
(4) Hepatitis, Haemophilia, AIDS

157. An ideal contraceptive should **not** be
(1) User-friendly
(2) Easy available
(3) Non-reversible
(4) Effective against pregnancy

158. Select the **odd** one w.r.t. barrier methods of contraception.
(1) Condom (2) Vault
(3) Diaphragm (4) Combined pills

159. Read the following statement (A) and statement (B) carefully and choose the **correct** option.
Statement (A) : Lippes loop is an effective contraceptive device.
Statement (B) : Lippes loop is a barrier method which prevents the deposition of sperms into the vagina of female.
(1) Both the statements are correct
(2) Both the statements are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct

160. Test tube baby does not differ from normally produced baby w.r.t.
(1) Mode of fertilisation
(2) Parturition
(3) *In-vitro* growth of zygote upto few celled stage
(4) Embryo transfer

161. According to MTP (amendment) act of which year in India, a pregnancy may be terminated on certain grounds within the first 12 weeks on the opinion of one registered medical practitioner?
(1) 1971 (2) 1951
(3) 2011 (4) 2017

162. Which of the following factors has **not** contributed to population explosion in India since independence?
(1) Rise in MMR
(2) Decline in IMR
(3) Increase in number of people in reproductive age group
(4) Rapid decline in death rate

163. The test tube baby programme does **not** include which one of the following techniques?
(1) GIFT (2) ZIFT
(3) ICSI (4) IUT

164. Oral contraceptive pills generally contain combination of oestrogen and progestogens. In females, probably these pills inhibit/alter which of the following?
a. LH and FSH secretion
b. Follicular development and ovulation
c. Quality of cervical mucus
Select the correct option
(1) a only (2) a and b only
(3) b and c only (4) a, b and c

165. All of the given diseases are completely curable if detected early and treated properly, **except**
(1) Syphilis (2) Gonorrhea
(3) Genital herpes (4) Trichomoniasis

166. Contraceptive method that is generally advised for the female partner as terminal method to prevent any more pregnancies is
(1) Vasectomy (2) Tubectomy
(3) Oophorectomy (4) Hysterectomy

167. One-fifth of the total number of conceived pregnancies are lost every year, all over the world, due to
(1) Failure of sterilisation operations
(2) Complications of STDs
(3) Voluntary termination
(4) Weak health of mothers

168. Lactational amenorrhoea is a natural method of birth control but is generally effective upto
(1) 8 months after parturition (2) 6 months after parturition
(3) 2 years after parturition (4) 1 year after parturition

169. All of the following are steps taken by government to check the increasing population growth, **except**
(1) Motivating smaller families and to adopt various contraceptive methods
(2) Raising of marriageable age of the female to 18 years and that of males to 21 years
(3) Provision for incentives given to couples with small families
(4) Promotion of abortions, if second pregnancy occurs

170. A woman can provide a suitable enviroment for embryonic development and implantation but has blocked fallopian tubes, which is making natural conception almost impossible. If she still wishes to conceive, then which of the following ART methods would be recommended to her?
(1) IUI (2) ZIFT
(3) GIFT (4) IVF followed by IUT

171. Which of the following chemicals can be used under chemical methods for contraception?
(A) Lactic acid
(B) Boric acid
(C) Citric acid
Choose the correct option.
(1) (A) and (B) only (2) (B) and (C) only
(3) (A) and (C) only (4) (A), (B) and (C)

172. 'Nirodh' is a popular brand of
(1) IUD for female
(2) Contraceptive pill for female
(3) Condom for male
(4) Condom for female

173. The diaphragm is a rubber sheath having dome-shaped structure, and it stops the sperms from entering into
(1) Vestibule (2) Vagina
(3) Cervix (4) Copulation canal

174. Injectable contraceptives contain
(A) Progestogen alone
(B) Combination of progestogen and estrogen
(C) FSH and LH
Choose the **correct** option.
(1) (A), (B) and (C) (2) (B) and (C) only
(3) (A) and (C) only (4) (A) and (B) only

175. A birth control implant having six small silicone capsules and with the effective period upto 5 years is
(1) Norplant (2) Multiload-375
(3) Perceptin (4) Depo-Provera

176. Read the given statements and select the **correct** option.
Statement A: Spermicidal creams, jellies and foams are used along with some barrier contraceptive devices.
Statement B: Spermicidal creams decrease the contraceptive efficiency of barrier contraceptive devices.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

177. Which of the following has/have been found to be very effective as emergency contraceptive(s) as it/they could be used to avoid possible pregnancy due to rape, if given within 72 hours of coitus?
A. Progestogens
B. Progestogen-estrogen combination
C. IUDs
Choose the **correct** option.
(1) A only
(2) A and B only
(3) B only
(4) A, B and C

178. Castration cannot be taken as a contraceptive method, because
(1) Deficiency of testosterone hormone will deprive a person of sexual drive, desire or the sexual act
(2) It inhibits gamete transport
(3) It is a surgical method of contraception, also called sterilisation
(4) This technique is highly effective and its reversibility is high.

179. Among the following methods/devices, which one has the highest failure rate?
(1) Diaphragm used with spermicide
(2) Condom
(3) Intrauterine device
(4) Rhythm method

180. Which of the following STIs is caused by the human papilloma virus (HPV) and is transmitted through intimate contact with an infected person?
(1) Genital herpes
(2) Genital warts
(3) AIDS
(4) Chlamydiasis

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